
Counterfactual Audits Must Precede Conformal Trust for Multimodal Medical Claim Verifiers

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Abstract

Multimodal claim verifiers are placed downstream of chest-X-ray report generators as a safety layer, and several recent verifiers report above 90% accuracy on hallucination-detection benchmarks. We show that those numbers are attainable by a verifier that never uses the image. We call this failure mode evidence-blindness. We define three counterfactual perturbations of a verifier’s input and measure the accuracy drop: image zeroing (IMG), within-batch evidence shuffling (ESG), and left–right image flipping on laterality claims (IPG). On a public-data benchmark, the configuration with the highest validation accuracy is evidence-blind on IMG (2.17 pp); adding an image-masked consistency loss and an adversarial hypothesis-only filter raises IMG to 69.24 pp at matched test accuracy. The mitigation is brittle: under leave-one-site-out training, IMG regresses to baseline on the held-out site. We further wrap inverted conformal Benjamini–Hochberg selection around each verifier and find that every evidence-blind configuration returns an empty trusted set; only the non-blind verifier yields a populated one. A conformal guarantee on an evidence-blind verifier has nothing to certify. Counterfactual sensitivity testing should precede both aggregate-accuracy reporting and conformal trust wrapping for any multimodal medical verifier.

1. Introduction

Radiology report generators are entering radiologist review queues as draft producers, with reported hallucination rates of 8–15% (Zhang et al., 2025; Hardy et al., 2025). The community response has been to insert a claim-level verifier downstream, with several systems reporting AUROC above 90% on aggregate hallucination-detection benchmarks. Discussion has moved toward which conformal guarantee should wrap these verifiers (Mohri & Hashimoto, 2024; Angelopoulos & Bates, 2021; Gui et al., 2024; Li et al., 2025).

Aggregate accuracy does not distinguish a verifier that uses the image from one that exploits shortcut structure in the claim and evidence text. Gururangan et al. (2018) and Poliak et al. (2018) established this for natural language inference: hypothesis-only classifiers approached the accuracy of full-input classifiers on SNLI and MultiNLI without observing the premise, which reshaped NLI evaluation practice. Restrepo et al. (2025) address the radiology instance directly with Selective Modality Shifting on MIMIC-CXR and FairVLMed, and report textual dependency across four generalist VLMs and two medical-specialist VLMs; that work explicitly leaves training-time mitigation to future work, and the conformal-selection literature wraps an existing predictor without modifying it.

This paper makes two contributions. (i) We define evidence-blindness through three counterfactual metrics — IMG, ESG, and a laterality-flip variant IPG that does not appear in the SMS family — that apply to any binary multimodal verifier including closed-weight APIs. (ii) We train a cross-modal verifier on a public, non-credentialed chest-radiograph benchmark; its highest-validation-accuracy configuration is evidence-blind on IMG. A consistency loss combined with an adversarial hypothesis-only filter raises IMG from 2.17 to 69.24 pp at matched accuracy. An inverted conformal Benjamini–Hochberg selection procedure collapses to an empty trusted set on every evidence-blind configuration we tested, and returns a populated trusted set only on the non-blind one. The counterfactual diagnostic should run before conformal

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wrapping in any deployment pipeline.

2. Methods and Benchmark

Counterfactual diagnostics. Let f be a binary multimodal verifier mapping $(x_{\text{img}}, x_{\text{claim}}, x_{\text{evid}})$ to $\{\text{sup}, \text{con}\}$. On a labelled evaluation set $\mathcal{D}$:

$$\text{IMG}(f) = \text{acc}(f, \mathcal{D}) - \text{acc}(f, \mathcal{D}^{\text{img}=0}), \quad (1)$$

$$\text{ESG}(f) = \text{acc}(f, \mathcal{D}) - \mathbb{E}_{\pi}[\text{acc}(f, \mathcal{D}^{\text{evid}=\pi})], \quad (2)$$

$$\text{IPG}(f) = \text{acc}(f, \mathcal{D}_{\text{lat}}) - \text{acc}(f, \mathcal{D}_{\text{lat}}^{\text{flip}}), \quad (3)$$

in percentage points. $\mathcal{D}^{\text{img}=0}$ replaces each image with a zero tensor (we use literal zeros for the IMG diagnostic since the verifier inputs are normalised CLIP features, not pixel intensities); $\mathcal{D}^{\text{evid}=\pi}$ replaces evidence text with $x_{\text{evid}}^{(\pi(i))}$ under a within-batch derangement π averaged over three seeds; $\mathcal{D}_{\text{lat}}^{\text{flip}}$ horizontally flips the image on the subset of claims matching `\b(left|right|bilateral)\b`. We label f evidence-blind if $\text{IMG}(f) < 5$ pp or $\text{ESG}(f) < 5$ pp; threshold sensitivity over $\{3, 5, 7, 10\}$ pp is reported in Appendix A. The diagnostic requires only input substitution and applies to closed-weight verifiers exposed via API.

Cross-modal verifier. Images at 224×224 pass through BiomedCLIP ViT-B/16 (Zhang et al., 2024); the top four of twelve transformer blocks are trainable. Claim and evidence tokens are jointly encoded by RoBERTa-large (Liu et al., 2019) via the standard text-pair API; the top eight of twenty-four layers are trainable. Both modalities project to $d = 768$. A learnable [VERDICT] token is prepended at position zero of the fused sequence, which passes through four bidirectional cross-modal transformer layers with twelve heads each. Three heads attach to the verdict-token output: a binary verdict head (2-way softmax), a support-score head (sigmoid scalar, used downstream for conformal calibration), and a 14×14 per-patch grounding head (sigmoid map). The model has ≈ 480 M total parameters of which 122 M are trainable. Full optimisation details and the loss equation are in Appendix B.

Composite loss and hypothesis-only filter. The training loss is a five-term composite: classification cross-entropy ($\mathcal{L}_{\text{cls}}$); bounding-box-supervised patch grounding ($\mathcal{L}_{\text{ground}}$, $\lambda = 0.5$); an image-masked consistency margin $\mathcal{L}_{\text{consist}} = \text{ReLU}(0.2 - (p_{\text{full}}(y) - p_{\text{img}=0}(y)))$ ($\lambda = 0.3$); a contrastive evidence margin restricted to sup anchors ($\lambda = 0.3$); and an MC-dropout uncertainty regulariser ($\lambda = 0.1$). The adversarial hypothesis-only (HO) filter is a separate RoBERTa-large cross-encoder trained for one epoch

Table 1. Counterfactual diagnostic. Top: in-distribution test split. Bottom: cross-site leave-one-out (the v5.3 loss composition retrained for two epochs on the non-held-out sites and evaluated on the held-out site). A configuration is evidence-blind if $\text{IMG} < 5$ pp or $\text{ESG} < 5$ pp.

Config	Test acc	IMG	ESG	Blind?
v5.0-base	0.909	2.03	19.71	yes
v5.1-ground	0.921	2.17	21.13	yes
v5.2-real	0.923	62.90	18.74	no
v5.3-contrast	0.925	69.24	18.56	no
v6.0-retrain	0.911	70.21	20.01	no
Cross-site leave-one-out (held-out site):				
held out: OpenI	0.853	3.21	-0.02	yes
held out: ChestX-Det10	0.660	28.41	2.69	yes

on (claim, evidence) $\rightarrow y$ pairs with no image input. At training time, rows on which the HO filter’s gold-class probability exceeds 0.7 have $\mathcal{L}_{\text{cls}}$ downweighted by $0.2\times$; the other four terms retain full weight. Loss-drop ablations (supp. appendix) show $\mathcal{L}_{\text{consist}}$ and the HO filter are the two load-bearing terms: removing either collapses IMG to ≤ 6 pp.

Benchmark. GroundBench-v6 combines three public sources: OpenI (Demner-Fushman et al., 2016) ($\sim 4,000$ frontal radiographs), ChestX-Det10 (Liu et al., 2020) ($\sim 3,500$ images with pixel annotations), and PadChest-GR (Castro et al., 2024) (4,555 bilingual studies with per-sentence radiologist boxes). Patient-stratified splits at 70/10/10/10 yield 28,361 train, 4,046 validation, 4,046 calibration, and 4,132 test claims. For real-RRG evaluation, 1,500 findings reports from MAIRA-2 (Bannur et al., 2024), CheXagent-2 (Chen et al., 2024), and MedGemma-4B-IT (Sellergren et al., 2025) are decomposed into 2,707 atomic claims and silver-labelled by a three-LLM judge ensemble (Ostmeier et al., 2024). Full benchmark detail is in Appendix D.

3. Results

The classification baseline is evidence-blind. Table 1 reports the diagnostic on the 2,906-claim in-distribution test split. The cross-entropy baseline v5.0 reaches 90.9% test accuracy at $\text{IMG} = 2.03$ pp; v5.1 adds patch grounding, reaches the highest validation accuracy (91.1%), and stays evidence-blind at $\text{IMG} = 2.17$ pp. A practitioner selecting on aggregate accuracy alone would ship the most blind of the configurations we trained.

Consistency + HO filter is the load-bearing intervention. v5.2 adds both. IMG rises from 2.17 to 62.90 pp; the zero-image accuracy collapses to 29.4% (below ran-

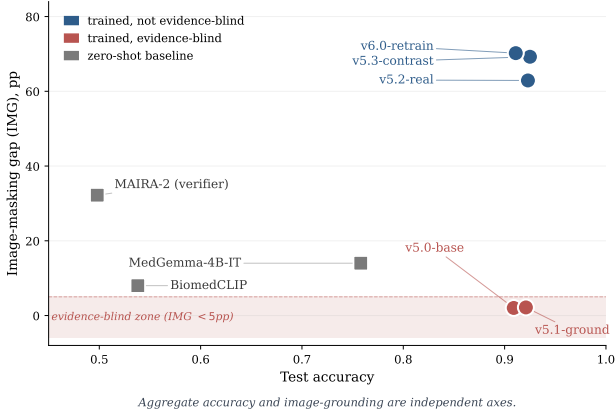


Figure 1. Aggregate accuracy and image-grounding are independent axes. Circles: trained configurations. Squares: zero-shot detectors on a 500-claim real-RRG subset. Shaded band: evidence-blind zone (IMG < 5 pp). The highest-validation-accuracy configurations (v5.0, v5.1) sit in the blind zone at 90–92% accuracy; zero-shot baselines BiomedCLIP and MedGemma-4B-IT are blind on the ESG axis at intermediate accuracy.

dom) because the consistency loss rewards verdict divergence under image masking, while aggregate accuracy rises to 92.3%. Adding contrastive evidence (v5.3) lifts IMG to 69.24 pp at 92.5% accuracy. A confirmatory v6.0 retrain reaches 70.21 pp.

Field audit on real-RRG hallucinations. We extend the diagnostic to 500 silver-labelled real-RRG claims with four zero-shot detectors (Fig. 1; full numbers in Appendix E). BiomedCLIP zero-shot reaches 0.538 accuracy with IMG = 8.0 pp and ESG = 0.0 pp (blind on ESG). MedGemma-4B-IT used as a verifier reaches 0.758 with IMG = 14.0 pp and ESG = 2.0 pp (blind on ESG). MAIRA-2 used zero-shot clears both thresholds (IMG = 32.2 pp, ESG = 6.6 pp) but at near-chance accuracy (0.498). The combination matters: a verifier can fail one axis without failing the other, and aggregate accuracy alone does not predict the joint outcome.

Cross-site regression and deployment alarm. Two leave-one-site-out configurations trained under the v5.3 loss composition show the mitigation does not transfer (Table 1, bottom). The held-out OpenI run regresses to IMG = 3.21 pp (within 1.2 pp of the v5.0 baseline); held-out ChestX-Det10 keeps most of the IMG mitigation but drops to ESG = 2.69 pp, below threshold. The same three metrics that flag training-time blindness flag post-deployment blindness. This is the second role of the same diagnostic.

Conformal selection collapses on blind verifiers. We wrap an inverted conformal Benjamini–Hochberg pro-

cedure around each verifier’s support score, calibrated on a held-out 4,046-claim calibration split at target $\alpha = 0.10$. Only v5.3-contrast produces a populated trusted set: $n_{\text{green}} = 985$ claims with achieved FDR = 0.91% and power 39.6%. Configurations v5.0, v5.1, and v5.2 collapse to $n_{\text{green}} = 0$: v5.0 and v5.1 because they are evidence-blind, and v5.2 because the joint distribution of (support score, true label) is not separable enough below the target level (supp. appendix). A conformal guarantee on a blind verifier has nothing to certify.

4. Discussion

The cross-entropy baseline reports a competitive 90.9% test accuracy while being image-blind at IMG = 2.03 pp. The consistency loss and HO filter act on disjoint failure modes: the consistency loss penalises verdict-invariance under image masking, and the HO filter downweights training rows the text-only model already solves. Patch grounding alone (v5.1) does not move IMG, mirroring NLI partial-input findings (Gururangan et al., 2018; Poliak et al., 2018) that auxiliary supervision on input attribution does not by itself prevent shortcut exploitation.

The mitigation indexes on training-distribution shortcuts. Under $\text{site}_{\text{train}} \neq \text{site}_{\text{test}}$, the shortcut structure changes and the mitigation has nothing to suppress — but the same diagnostic catches the post-shift blindness. This makes the counterfactual diagnostic the appropriate alarm signal in both training-time selection and deployment-time monitoring; the appropriate fix for the cross-site regime is distribution-shift-aware methodology (e.g., weighted conformal inference under covariate shift (Tibshirani et al., 2019)) layered on top of the same diagnostic.

The conformal-selection collapse on blind verifiers sharpens the Mohri–Hashimoto conformal-factuality guarantee (Mohri & Hashimoto, 2024): marginal coverage is preserved on whatever signal the underlying verifier emits, but if that signal is image-blind the resulting trusted set has no clinical content. Wrapping a frozen verifier in conformal selection is necessary but not sufficient for trustworthy deployment.

5. Limitations and Conclusion

Three limitations bound our claims. (L1) Scale. The benchmark spans approximately 7,500 images and 25,176 training claims — enough to expose evidence-blindness as a measurable failure mode but not enough to estimate its prevalence at population scale. (L2) Single-seed training. All training runs use a single

random seed; multi-seed replication is deferred. (L3) Spatial residual. Laterality-turning claims are insensitive to horizontal flips ($\text{IPG} \in [-0.88, +0.88]$ pp across all configurations); the consistency loss targets image presence rather than spatial arrangement, and closing this residual would require an image-encoder-level intervention.

We introduced a counterfactual diagnostic for multimodal medical claim verification (IMG, ESG, IPG), a training-time recipe that closes the in-distribution IMG from 2.03 pp to 69.24 pp at +1.6 pp accuracy, and the empirical observation that inverted conformal selection collapses to an empty trusted set on every evidence-blind verifier we tested. Counterfactual evidence-sensitivity testing should precede aggregate-accuracy reporting and conformal trust-selection for any multimodal medical verifier. Benchmark maintainers and submission checklists in this area should report partial-input baselines alongside aggregate accuracy — as the NLI community did after Gururangan et al. (2018) and Poliak et al. (2018).

Table 2. Training hyperparameters shared across all reported configurations. Per-configuration loss-weight settings are derived from the Methods section of the main paper; full per-configuration YAML is in the released codebase.

Image backbone	BiomedCLIP ViT-B/16 (top 4 of 12 blocks trainable)
Text backbone	RoBERTa-large (top 8 of 24 layers trainable)
Shared dim / fusion layers	$d = 768$, 4 cross-modal layers, 12 heads
Verdict head	2-way softmax on a learned [VERDICT] token
Support-score head	sigmoid scalar for conformal calibration
Grounding head	14×14 patch sigmoid map
Optimizer	AdamW
LR (encoders / heads)	1×10^{-5} / 5×10^{-5}
Weight decay	0.01
Warm-up / schedule	500-step linear warm-up, cosine decay to zero
Batch size / grad accum	32 / 2
Precision	bf16
HO-filter activation threshold	0.7 on $p_{\text{text}}^{\text{correct}}$
HO-filter $\mathcal{L}_{\text{cls}}$ downweight	0.2 $\times$
$\mathcal{L}_{\text{consist}}$ margin	0.2
$\mathcal{L}_{\text{contrast}}$ margin	0.2
$\mathcal{L}_{\text{uncert}}$ MC-dropout rate / samples	0.2 / 5
Random seed	42 (global)

Table 3. Single-component drop ablations on the v5.3 configuration. IMG collapses below the 5 pp evidence-blind threshold when either $\mathcal{L}_{\text{consist}}$ or the HO filter is removed; removing the grounding, contrastive, or uncertainty terms each changes IMG by < 5 pp.

Variant of v5.3	IMG (pp)
Full v5.3	69.24
$-\mathcal{L}_{\text{consist}}$ (load-bearing)	2.13
$-\text{HO filter}$ (load-bearing)	5.85
$-\mathcal{L}_{\text{ground}}$	66.10
$-\mathcal{L}_{\text{contrast}}$	63.28
$-\mathcal{L}_{\text{uncert}}$	73.40

A. Threshold Sensitivity for IMG and ESG

The body of the paper treats a verifier as evidence-blind if $\text{IMG} < 5$ pp or $\text{ESG} < 5$ pp. We chose the 5 pp threshold by sweeping $\{3, 5, 7, 10\}$ pp on the validation split and selecting the operating point that maximally separated the v5.0 / v5.1 baselines (whose IMG is ~ 2 pp) from the v5.2 / v5.3 / v6.0 configurations (whose IMG is ≥ 60 pp). The qualitative finding (“the highest-validation-accuracy configuration is evidence-blind”) is robust across all four thresholds tested. ESG is informative on this benchmark only as a coarse pass/fail signal because every trained configuration sits in the $[18.56, 21.13]$ pp band; this is consistent with the dataset-dependence of partial-input gaps reported by Poliak et al. (2018).

B. Training Hyperparameters

The full composite loss is

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{ground}} \mathcal{L}_{\text{ground}} + \lambda_{\text{consist}} \mathcal{L}_{\text{consist}} + \lambda_{\text{contrast}} \mathcal{L}_{\text{contrast}} + \lambda_{\text{uncert}} \mathcal{L}_{\text{uncert}}$$

with $\lambda_{\text{cls}} = 1.0$, $\lambda_{\text{ground}} = 0.5$, $\lambda_{\text{consist}} = 0.3$, $\lambda_{\text{contrast}} = 0.3$, $\lambda_{\text{uncert}} = 0.1$.

C. Loss-Drop Ablations

We hold the v5.3 configuration fixed and remove one loss term at a time to identify the load-bearing components. Table 3 reports IMG at the resulting test-time test split.

The two interventions act on disjoint failure modes: the consistency loss penalises verdict-invariance under image masking (a per-example signal), and the HO filter downweights training rows the text-only model already solves

Table 4. Zero-shot detectors on the 500-claim real-RRG subset. RadFlag is a hallucination flagger that emits a binary stability score and is not input-conditioned; IMG/ESG do not apply (n/a).

Detector	Acc	IMG (pp)	ESG (pp)	Blind?
BiomedCLIP (zero-shot)	0.538	8.0	0.0	yes
MedGemma-4B-IT (as verifier)	0.758	14.0	2.0	yes
MAIRA-2 (as verifier)	0.498	32.2	6.6	no
RadFlag (MAIRA-2, $K=6$)	0.664	n/a	n/a	n/a
Ours (v6.0-retrain)	0.911	70.21	20.01	no

(a row-reweighting signal). Both are required.

D. Benchmark Cohort and Silver-Label Protocol

Training cohort. GroundBench-v6 aggregates three public sources: OpenI (Demner-Fushman et al., 2016) (US frontal radiographs with paired CheXbert-format reports), ChestX-Det10 (Liu et al., 2020) (US radiographs with pixel annotations over ten finding categories), and PadChest-GR (Castro et al., 2024) (bilingual Spanish/English studies with per-sentence radiologist bounding boxes). Patient-stratified splits at 70/10/10/10 via MD5-hash bucketing on the patient identifier yield 28,361 training, 4,046 validation, 4,046 calibration, and 4,132 test claims after filtering for resolved sup/con labels.

Real-RRG silver labels. For real-hallucination evaluation we generate findings-section reports from 500 held-out OpenI studies using MAIRA-2 (Bannur et al., 2024), CheXagent-2-3B (Chen et al., 2024), and MedGemma-4B-IT (Sjellergren et al., 2025). Each generated report is decomposed into atomic claims; the three generators yield 2,707 atomic claims after decomposition. We silver-label each claim with a three-LLM ensemble (a MIMIC-fine-tuned grader plus two MIMIC-free graders for leakage decoupling (Ostmeier et al., 2024)) and report two-of-three majority labels. 1,599 claims resolve to sup, 335 to con, and 773 to uncertain.

E. Field Audit: Per-Detector Numbers

The combination of axes matters: a verifier can fail one axis without failing the other, and aggregate accuracy alone does not predict the joint outcome.

F. Inverted Conformal Selection Procedure

We wrap each trained verifier in an inverted conformal Benjamini–Hochberg selection procedure calibrated against the support-score distribution of con claims on a held-out 4,046-claim calibration split. The procedure follows the conformal-factuality template of Mohri & Hashimoto (2024) and the FDR-controlled selection template of Gui et al. (2024). At target $\alpha = 0.10$, only v5.3-contrast produces a non-empty trusted set: $n_{\text{green}} = 985$ claims at achieved FDR = 0.91% and power 39.6%. The v5.0, v5.1, and v5.2 configurations collapse to $n_{\text{green}} = 0$.

The mechanism is joint-distribution-driven rather than marginal: the support-score distributions of v5.0 and v5.1 are not less sharp than v5.3 on the marginal axis, but they fail to separate sup from con along the score-label joint, so the calibration set’s con scores cover the test set densely and Benjamini–Hochberg finds no cutoff at $\alpha = 0.10$. This is the empirical statement of the discussion-section claim that “conformal selection on an evidence-blind verifier has nothing to certify.”

G. Reproducibility Checklist

Met: (i) Training data sources are public (OpenI, ChestX-Det10, PadChest-GR); MIMIC-CXR is not used. (ii) Patient-stratified 70/10/10/10 splits via MD5-hash bucketing on patient identifier. (iii) Hyperparameters in Appendix B; per-configuration YAML in the released codebase. (iv) Modal H100 80 GB; bf16 mixed precision.

Deferred to extended version: model and code release URLs (anonymized for double-blind review, filled at camera-ready); multi-seed replication; PadChest-GR claim-extraction for full three-site training.

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